

Have you learned how to drive an F1 car by watching Checo Pérez race on TV?

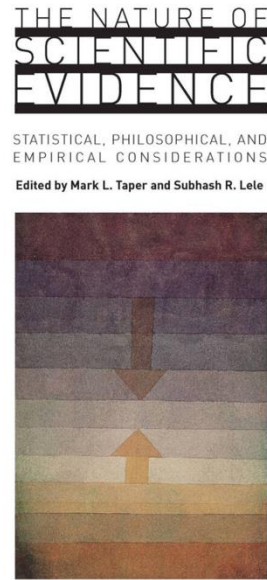
- If you only listen to lectures and are not actively reading, engaging with content, and making notes, you will not learn
- Practice many times the computer tutorials
- Find others in the class who can discuss ideas with you
- Seek for help early!



Introduction to the likelihood function

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MOLE at Woods Hole workshop 2026

**Purely
likelihood-
based
inference is
also known as
Evidential
statistics**



A branch of statistics interested in understanding how to infer processes through evidence (data)

Goal: To propose a way to measure evidence connecting it to hypotheses

Our example today: Bird alarm calls in the Peruvian Amazonia

Ari goes to the jungle and
observes the behavior of two
mixed flocks.

Dead-leaf gleaners feeding on the
ground would freeze after an
alarm call.

Flycatchers feeding on the top of
the trees would resume eating
almost instantly after an alarm
call.



Ari Martinez
UC Santa Cruz



Thamnomanes ardesiacus





The experiment and gathering the data

What do you notice about these data?

Response	Forage
18	DL
11	DL
23	DL
0	F
0	F
6	F

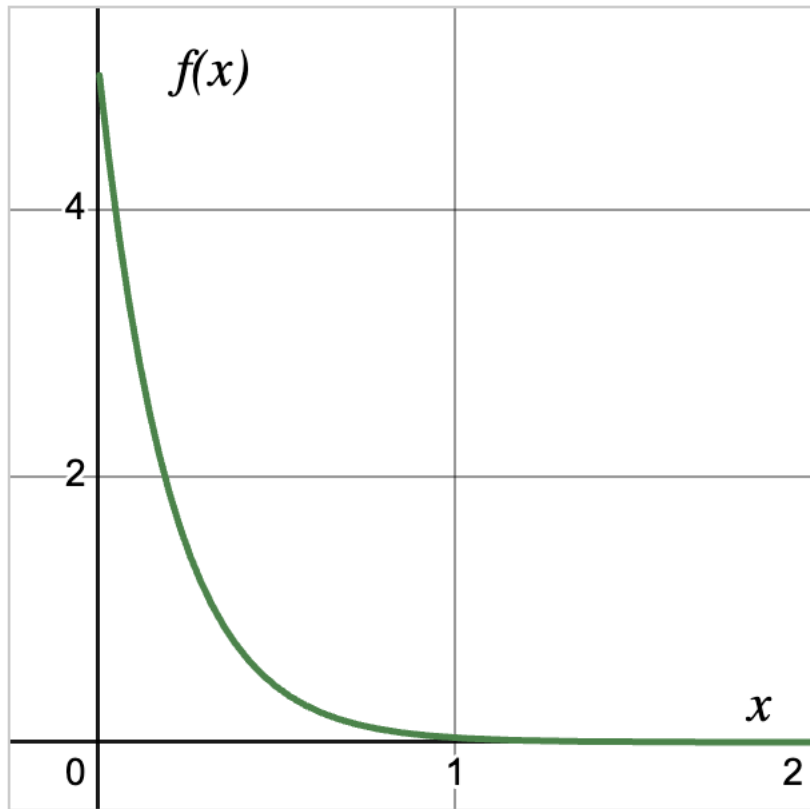
Definition of a random variable

- A random variable is a function that takes us from the space of events (Ω) to a number, in this case a continuous number ($\mathbb{R}$)

Probability spaces and measure

$$(\Omega, \sigma - algebra, \mathbb{P}) \rightarrow (\mathbb{R}, B, P)$$

Probability model for freezing (waiting) times



Likelihood function (flycatchers only)

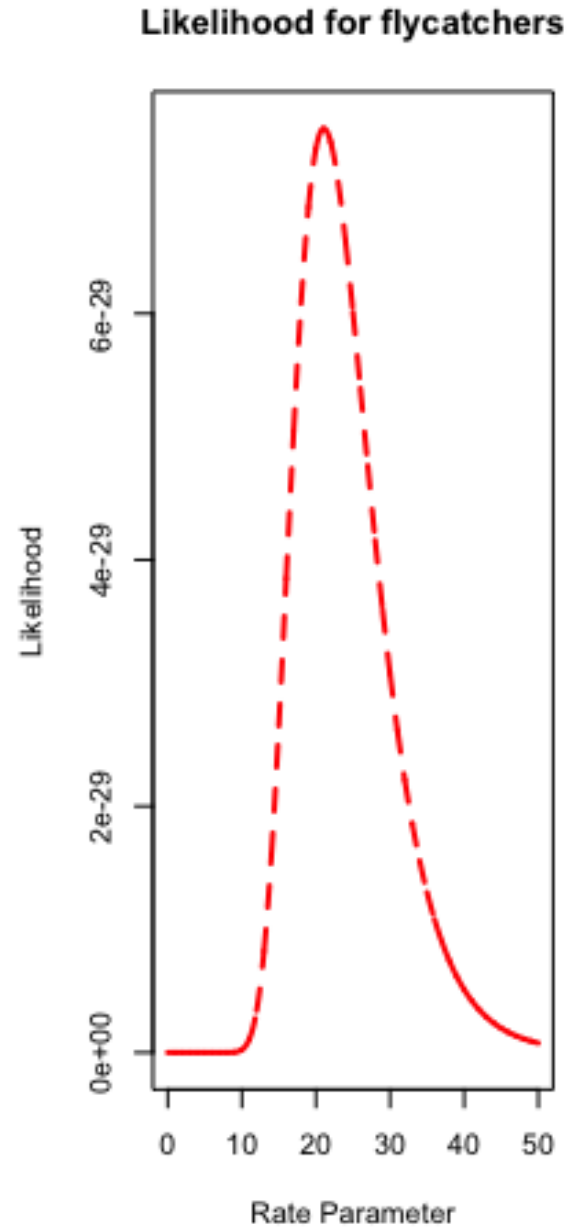
$$\mathcal{L}(\beta; \mathbf{X}) = P(\mathbf{X}|\beta)$$

The probability of the sample given the model or hypothesis represented by the symbol θ

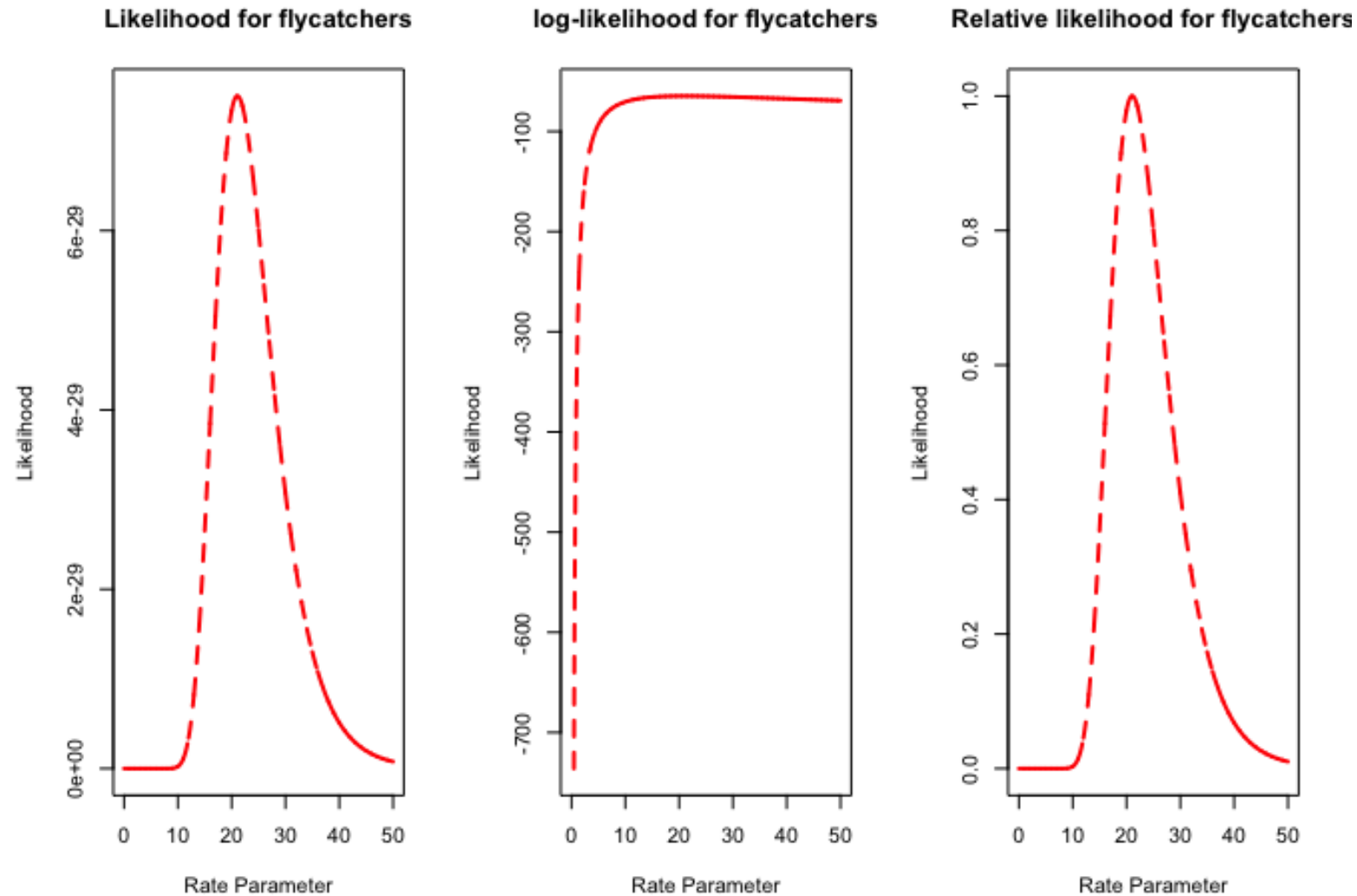
1. The model:
2. Data are assumed independent:

Calculating the likelihood function

Maximum likelihood estimate (MLE)



Different ways of visualizing the likelihood

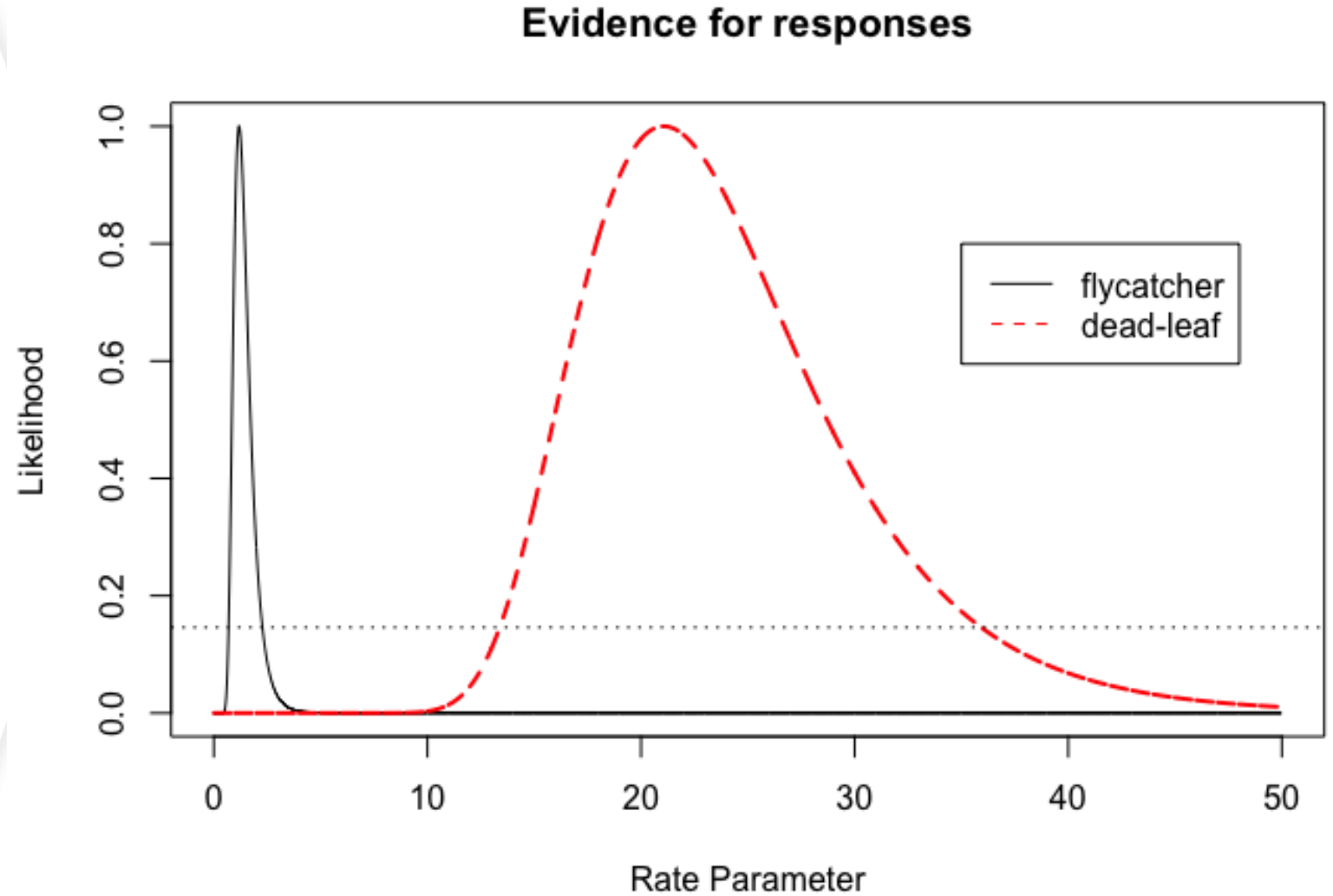


Exercise (in class if we have time)

- Calculate the likelihood for the dead-leaf gleaners:

Data: $X = \{18, 11, 23\}$ for parameter values $\beta = 15, 16, 17, 18$

Differences between the groups



Advantages of using a likelihood function

(Sprott and Kalbfleisch, 1965)

1. It represents a concise summary of all of the information in the data, and when graphed, it gives a pictorial summary of what the data have to say.
2. Inferences based on the likelihood are exact for samples of all sizes; no assumption or mathematical investigation is required to ascertain if the sample is large enough to justify the method.
3. The rather complicated mathematics of asymptotic or large sample theory is not necessary, giving a relatively simple theoretical approach to inference.

Disadvantages

(Sprott and Kalbfleisch, 1965)

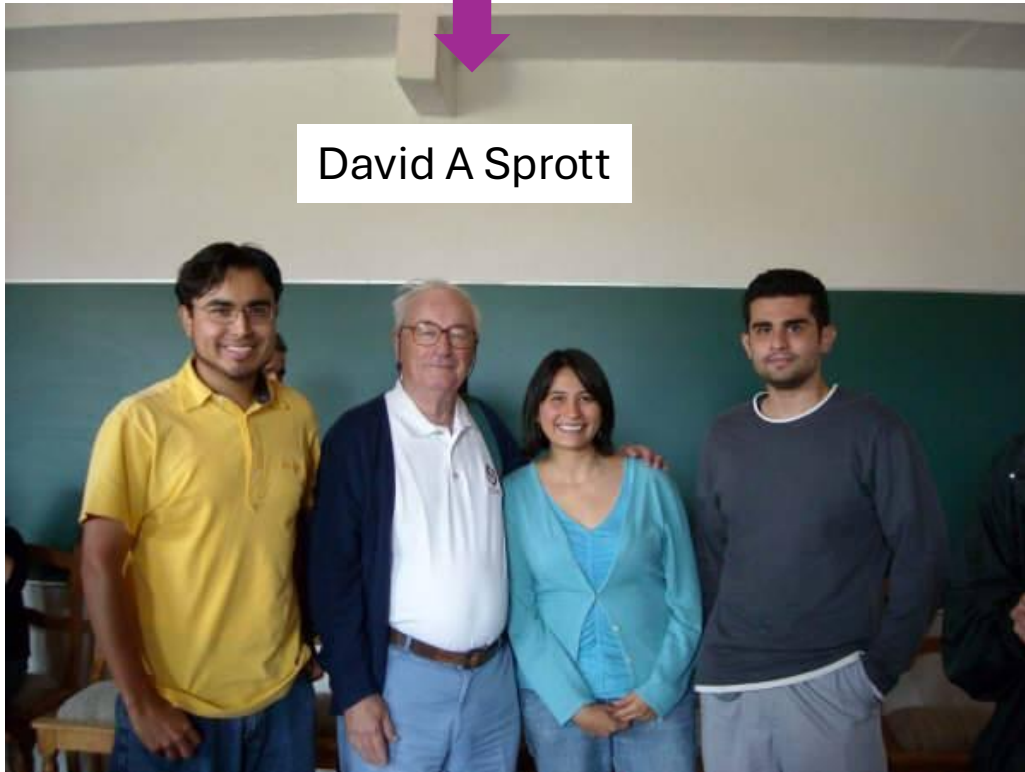
1. Increased numerical computation in addition to the uncertainty being expressed (exactly) in terms of likelihood rather than (approximately in large samples) in terms of probability.

2. The likelihood method is not a test of significance and cannot be used to test the validity of a single hypothesis without reference to alternatives

Fisher



David A Sprott



Collective dynamics of 'small-world' networks

Duncan J. Watts* & Steven H. Strogatz

